



**中认信通**

CHINA CERTIFICATION ICT CO., LTD (DONGGUAN)



# TEST REPORT

**Applicant:** Dragino Technology Co., Limited.

**Address:** Room 202, BaoChengTai industrial park, No.8 CaiYun LongCheng Street, LongGang District, Shenzhen China

**FCC ID:** ZHZ485LB

**Product Name:** LoRaWAN Sensor Node

**Standard(s):** 47 CFR Part 15, Subpart C(15.247)  
ANSI C63.10-2020  
KDB 558074 D01 15.247 Meas Guidance v05r02

The above device has been tested and found compliant with the requirement of the relative standards by China Certification ICT Co., Ltd (Dongguan)

**Report Number:** CR230844753-RF-00CA1

**Date Of Issue:** 2025/10/8

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**Declarations**

China Certification ICT Co., Ltd (Dongguan) is not responsible for the authenticity of any test data provided by the applicant. Data included from the applicant that may affect test results are marked with a triangle symbol “▲”. Customer model name, addresses, names, trademarks etc. are not considered data.

Unless otherwise stated the results shown in this test report refer only to the sample(s) tested.

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This report may contain data that are not covered by the accreditation scope and shall be marked with an asterisk “★”.

Each test item follows the test standard(s) without deviation.

# CONTENTS

|                                                                     |           |
|---------------------------------------------------------------------|-----------|
| <b>DOCUMENT REVISION HISTORY .....</b>                              | <b>5</b>  |
| <b>1. GENERAL INFORMATION .....</b>                                 | <b>6</b>  |
| <b>1.1 PRODUCT DESCRIPTION FOR EQUIPMENT UNDER TEST (EUT) .....</b> | <b>6</b>  |
| 1.1.1 General.....                                                  | 6         |
| 1.1.2 Operation Frequency Detail .....                              | 7         |
| 1.1.3 Antenna Information Detail <sup>▲</sup> .....                 | 7         |
| 1.1.4 Accessory Information.....                                    | 7         |
| <b>1.2 DESCRIPTION OF TEST CONFIGURATION.....</b>                   | <b>8</b>  |
| 1.2.1 EUT Operation Condition:.....                                 | 8         |
| 1.2.2 Support Equipment List and Details .....                      | 8         |
| 1.2.3 Support Cable List and Details .....                          | 8         |
| 1.2.4 Block Diagram of Test Setup .....                             | 8         |
| <b>1.3 MEASUREMENT UNCERTAINTY .....</b>                            | <b>9</b>  |
| <b>2. SUMMARY OF TEST RESULTS .....</b>                             | <b>10</b> |
| <b>3. REQUIREMENTS AND TEST PROCEDURES .....</b>                    | <b>11</b> |
| <b>3.1 AC LINE CONDUCTED EMISSIONS.....</b>                         | <b>11</b> |
| 3.1.1 Applicable Standard.....                                      | 11        |
| 3.1.2 EUT Setup.....                                                | 12        |
| 3.1.3 EMI Test Receiver Setup .....                                 | 12        |
| 3.1.4 Test Procedure .....                                          | 13        |
| 3.1.5 Corrected Amplitude & Margin Calculation.....                 | 13        |
| <b>3.2 RADIATION SPURIOUS EMISSIONS .....</b>                       | <b>14</b> |
| 3.2.1 Applicable Standard.....                                      | 14        |
| 3.2.2 EUT Setup.....                                                | 14        |
| 3.2.3 EMI Test Receiver & Spectrum Analyzer Setup .....             | 15        |
| 3.2.4 Test Procedure .....                                          | 15        |
| 3.2.5 Corrected Amplitude & Margin Calculation.....                 | 15        |
| <b>3.3 6 DB EMISSION BANDWIDTH: .....</b>                           | <b>16</b> |
| 3.3.1 Applicable Standard.....                                      | 16        |
| 3.3.2 EUT Setup.....                                                | 16        |
| 3.3.3 Test Procedure .....                                          | 16        |
| <b>3.4 MAXIMUM CONDUCTED OUTPUT POWER: .....</b>                    | <b>17</b> |
| 3.4.1 Applicable Standard.....                                      | 17        |
| 3.4.2 EUT Setup.....                                                | 17        |
| 3.4.3 Test Procedure .....                                          | 17        |
| <b>3.5 MAXIMUM POWER SPECTRAL DENSITY: .....</b>                    | <b>18</b> |
| 3.5.1 Applicable Standard.....                                      | 18        |
| 3.5.2 EUT Setup.....                                                | 18        |
| 3.5.3 Test Procedure .....                                          | 18        |
| <b>3.6 100 KHZ BANDWIDTH OF FREQUENCY BAND EDGE: .....</b>          | <b>19</b> |
| 3.6.1 Applicable Standard.....                                      | 19        |
| 3.6.2 EUT Setup.....                                                | 19        |

|                                                     |           |
|-----------------------------------------------------|-----------|
| 3.6.3 Test Procedure .....                          | 19        |
| <b>3.7 DUTY CYCLE:.....</b>                         | <b>20</b> |
| 3.7.1 EUT Setup.....                                | 20        |
| 3.7.2 Test Procedure .....                          | 20        |
| <b>3.8 ANTENNA REQUIREMENT.....</b>                 | <b>20</b> |
| 3.8.1 Applicable Standard.....                      | 20        |
| 3.8.2 Judgment.....                                 | 20        |
| <b>4. Test DATA AND RESULTS .....</b>               | <b>21</b> |
| 4.1 AC LINE CONDUCTED EMISSIONS.....                | 21        |
| 4.2 RADIATION SPURIOUS EMISSIONS .....              | 22        |
| 4.3 6 dB EMISSION BANDWIDTH: .....                  | 28        |
| 4.4 MAXIMUM CONDUCTED OUTPUT POWER: .....           | 30        |
| 4.5 MAXIMUM POWER SPECTRAL DENSITY: .....           | 32        |
| 4.6 100 kHz BANDWIDTH OF FREQUENCY BAND EDGE: ..... | 34        |
| 4.7 DUTY CYCLE:.....                                | 36        |
| <b>5. RF EXPOSURE EVALUATION .....</b>              | <b>37</b> |
| 5.1 APPLICABLE STANDARD.....                        | 37        |
| 5.2 MEASUREMENT RESULT .....                        | 37        |
| <b>6. EUT PHOTOGRAPHS .....</b>                     | <b>38</b> |
| <b>7. TEST SETUP PHOTOGRAPHS .....</b>              | <b>39</b> |

**DOCUMENT REVISION HISTORY**

| Revision Number | Report Number        | Description of Revision | Date of Revision |
|-----------------|----------------------|-------------------------|------------------|
| 1.0             | CR230844753-00C      | Original Report         | 2023/10/7        |
| 2.0             | CR230844753-RF-00CA1 | Amended Report          | 2025/10/08       |

Note: This is the first amended report application which was based on the original report. The differences between them as following:

1. Added the models “LDS25-LB, LDS25-LS, LDS25-LB2, CS01-LB, CS01-LS, CS01-LB2, SEO1-LS, SEO1-LB2, DDS75-LS, DDS75-LB2, DDS45-LS, DDS45-LB2, DDS20-LS, DDS20-LB2, MDS120-LS, MDS120-LB2, SPH01-LS, SPH01-LB2, LMS01-LS, LMS01-LB2, WOS-LB, WQS-LS, WQS-LB2, SEOX-LB, SEOX-LS, SEOX-LB2, WSC2-LB, WSC2-LS, WSC2-LB2, MDS200-LS, MDS200-LB2, LDS40-LB, LDS40-LS, LDS40-LB2, LDS12-LS, LDS12-LB2, DDS04-LS, DDS04-LB2, RS485-LB2, RS485-LS, RS485W-LB, RS485W-LS, RS485W-LB2”.

The change between the previous equipment and the current equipment is stated and guaranteed by the applicant. The difference between them will not affect the test results, we will keep the test data, test photos and EUT photos.

## 1. GENERAL INFORMATION

### 1.1 Product Description for Equipment under Test (EUT)

#### 1.1.1 General

|                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>EUT Name:</b>                                                                                                                                                                                                     | LoRaWAN Sensor Node                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Trade Name:</b>                                                                                                                                                                                                   | DRAGINO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>EUT Model:</b>                                                                                                                                                                                                    | DDS75-LB                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Multiple Model(s):</b>                                                                                                                                                                                            | LDS25-LB, LDS25-LS, LDS25-LB2, CS01-LB, CS01-LS, CS01-LB2, SE01-LB, SE01-LS, SE01-LB2, DDS75-LS, DDS75-LB2, DDS45-LB, DDS45-LS, DDS45-LB2, DDS20-LB, DDS20-LS, DDS20-LB2, MDS120-LB, MDS120-LS, MDS120-LB2, SPH01-LB, SPH01-LS, SPH01-LB2, LMS01-LB, LMS01-LS, LMS01-LB2, WQS-LB, WQS-LS, WQS-LB2, SE0X-LB, SE0X-LS, SE0X-LB2, WSC2-LB, WSC2-LS, WSC2-LB2, MDS200-LB, MDS200-LS, MDS200-LB2, LDS40-LB, LDS40-LS, LDS40-LB2, LDS12-LB, LDS12-LS, LDS12-LB2, DDS04-LB, DDS04-LS, DDS04-LB2, RS485-LB, RS485-LB2, RS485-LS, RS485W-LB, RS485W-LS, RS485W-LB2 |
| <b>Operation Frequency:</b>                                                                                                                                                                                          | 903-914.2 MHz                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Maximum Peak Output Power (Conducted):</b>                                                                                                                                                                        | 14.37 dBm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Technique:</b>                                                                                                                                                                                                    | DTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Rated Input Voltage:</b>                                                                                                                                                                                          | DC3.6V from battery                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Serial Number:</b>                                                                                                                                                                                                | 299I-5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>EUT Received Date:</b>                                                                                                                                                                                            | 2023/8/1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>EUT Received Status:</b>                                                                                                                                                                                          | Good                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Note: The Multiple models are electrically identical with the test model. Please refer to the declaration letter for more detail, which was provided by manufacturer. All tests were performed with model: DDS75-LB. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

**1.1.2 Operation Frequency Detail**

| Channel | Frequency (MHz) | Channel | Frequency (MHz) |
|---------|-----------------|---------|-----------------|
| 1       | 903             | 5       | 909.4           |
| 2       | 904.6           | 6       | 911             |
| 3       | 906.2           | 7       | 912.6           |
| 4       | 907.8           | 8       | 914.2           |

Per section 15.31(m), the below frequencies were performed the test as below:

| Test Channel | Frequency (MHz) |
|--------------|-----------------|
| Lowest       | 903             |
| Middle       | 907.8           |
| Highest      | 914.2           |

**1.1.3 Antenna Information Detail<sup>^</sup>**

| Antenna Type | input impedance (Ohm) | Frequency Range | Antenna Gain |
|--------------|-----------------------|-----------------|--------------|
| Dipole       | 50                    | 902-928MHz      | 2 dBi        |

The Method of §15.203 Compliance:

☐ Antenna was permanently attached to the unit.

☒ Antenna use a unique type of connector to attach to the EUT.

☐ Unit was professionally installed, and installer shall be responsible for verifying that the correct antenna is employed with the unit.

**1.1.4 Accessory Information**

| Accessory Description | Manufacturer | Model | Parameters |
|-----------------------|--------------|-------|------------|
| /                     | /            | /     | /          |

## 1.2 Description of Test Configuration

### 1.2.1 EUT Operation Condition:

|                                                                                                                                 |                                                                                                    |        |         |
|---------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|--------|---------|
| <b>EUT Operation Mode:</b>                                                                                                      | The system was configured for testing in Engineering Mode, which was provided by the manufacturer. |        |         |
| <b>Equipment Modifications:</b>                                                                                                 | No                                                                                                 |        |         |
| <b>EUT Exercise Software:</b>                                                                                                   | serial port utility                                                                                |        |         |
| The software was provided by manufacturer. The maximum power was configured as below, that was provided by the manufacturer ▲ : |                                                                                                    |        |         |
| Test Modes                                                                                                                      | Power Level Setting                                                                                |        |         |
|                                                                                                                                 | Lowest                                                                                             | Middle | Highest |
| Lora-DTS                                                                                                                        | 8                                                                                                  | 8      | 8       |

### 1.2.2 Support Equipment List and Details

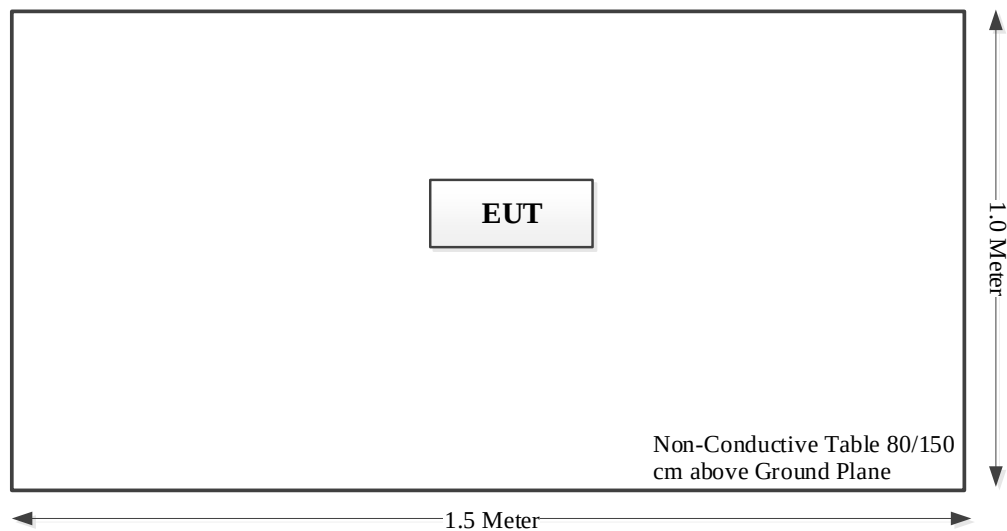
| Manufacturer | Description | Model | Serial Number |
|--------------|-------------|-------|---------------|
| /            | /           | /     | /             |

### 1.2.3 Support Cable List and Details

| Cable Description | Shielding Type | Ferrite Core | Length (m) | From Port | To |
|-------------------|----------------|--------------|------------|-----------|----|
| /                 | /              | /            | /          | /         | /  |

### 1.2.4 Block Diagram of Test Setup

Spurious emissions



### 1.3 Measurement Uncertainty

Otherwise required by the applicant or Product Regulations, Decision Rule in this report did not consider the uncertainty. The extended uncertainty given in this report is obtained by combining the standard uncertainty times the coverage factor K with the 95% confidence interval.

| Parameter                         | Measurement Uncertainty                                                                                           |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Occupied Channel Bandwidth        | ±5 %                                                                                                              |
| RF output power, conducted        | ±0.61dB                                                                                                           |
| Power Spectral Density, conducted | ±0.61 dB                                                                                                          |
| Unwanted Emissions, radiated      | 30M~200MHz: 4.15 dB,200M~1GHz: 5.61 dB,1G~6GHz: 5.14 dB,<br>6G~18GHz: 5.93 dB,18G~26.5G:5.47 dB,26.5G~40G:5.63 dB |
| Unwanted Emissions, conducted     | ±1.26 dB                                                                                                          |
| Temperature                       | ±1 °C                                                                                                             |
| Humidity                          | ±5%                                                                                                               |
| DC and low frequency voltages     | ±0.4%                                                                                                             |
| Duty Cycle                        | 1%                                                                                                                |
| AC Power Lines Conducted Emission | 2.8 dB (150 kHz to 30 MHz)                                                                                        |

## 2. SUMMARY OF TEST RESULTS

| Standard(s) Section                   | Test Items                               | Result         |
|---------------------------------------|------------------------------------------|----------------|
| §15.207(a)                            | AC line conducted emissions              | Not applicable |
| §15.205, §15.209,<br>§15.247(d)       | Spurious Emissions                       | Compliant*     |
| §15.247 (a)(2)                        | 6 dB Bandwidth                           | Compliant*     |
| §15.247(b)(3)                         | Maximum Conducted Output Power           | Compliant*     |
| §15.247(d)                            | 100 kHz Bandwidth of Frequency Band Edge | Compliant*     |
| §15.247(e)                            | Power Spectral Density                   | Compliant*     |
| §15.203                               | Antenna Requirement                      | Compliant*     |
| FCC §15.247 (i) & §1.1310&<br>§2.1091 | RF Exposure Evaluation                   | Compliant*     |

Note:

Compliant\*: The test data and related information was totally copied from the original report.

### 3. REQUIREMENTS AND TEST PROCEDURES

#### 3.1 AC Line Conducted Emissions

##### 3.1.1 Applicable Standard

FCC §15.207(a).

(a) Except as shown in paragraphs (b) and (c) of this section, for an intentional radiator that is designed to be connected to the public utility (AC) power line, the radio frequency voltage that is conducted back onto the AC power line on any frequency or frequencies, within the band 150 kHz to 30 MHz, shall not exceed the limits in the following table, as measured using a 50  $\mu$ H/50 ohms line impedance stabilization network (LISN). Compliance with the provisions of this paragraph shall be based on the measurement of the radio frequency voltage between each power line and ground at the power terminal. The lower limit applies at the boundary between the frequency ranges.

| Frequency of emission (MHz) | Conducted limit (dB $\mu$ V) |           |
|-----------------------------|------------------------------|-----------|
|                             | Quasi-peak                   | Average   |
| 0.15-0.5                    | 66 to 56*                    | 56 to 46* |
| 0.5-5                       | 56                           | 46        |
| 5-30                        | 60                           | 50        |

\*Decreases with the logarithm of the frequency.

(b) The limit shown in paragraph (a) of this section shall not apply to carrier current systems operating as intentional radiators on frequencies below 30 MHz. In lieu thereof, these carrier current systems shall be subject to the following standards:

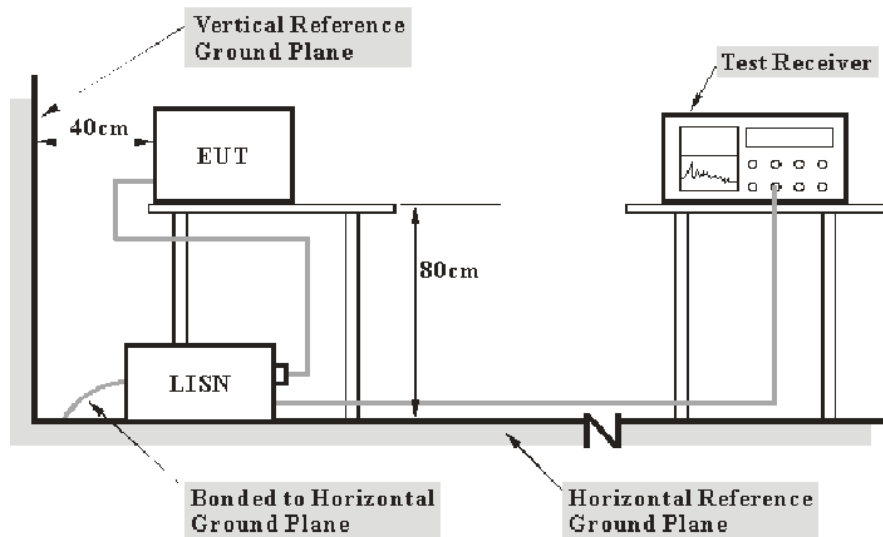
(1) For carrier current system containing their fundamental emission within the frequency band 535-1705 kHz and intended to be received using a standard AM broadcast receiver: no limit on conducted emissions.

(2) For all other carrier current systems: 1000  $\mu$ V within the frequency band 535-1705 kHz, as measured using a 50  $\mu$ H/50 ohms LISN.

(3) Carrier current systems operating below 30 MHz are also subject to the radiated emission limits in §15.205, §15.209, §15.221, §15.223, or §15.227, as appropriate.

(c) Measurements to demonstrate compliance with the conducted limits are not required for devices which only employ battery power for operation and which do not operate from the AC power lines or contain provisions for operation while connected to the AC power lines. Devices that include, or make provisions for, the use of battery chargers which permit operating while charging, AC adapters or battery eliminators or that connect to the AC power lines indirectly, obtaining their power through another device which is connected to the AC power lines, shall be tested to demonstrate compliance with the conducted limits.

### 3.1.2 EUT Setup



Note: 1. Support units were connected to second LISN.  
 2. Both of LISNs (AMN) 80 cm from EUT and at the least 80 cm from other units and other metal planes support units.

The setup of EUT is according with per **ANSI C63.10-2020** measurement procedure. The specification used was with the FCC Part 15.207 limits.

The spacing between the peripherals was 10 cm.

The adapter or EUT was connected to the main LISN with a 120 V/60 Hz AC power source.

### 3.1.3 EMI Test Receiver Setup

The EMI test receiver was set to investigate the spectrum from 150 kHz to 30 MHz.

During the conducted emission test, the EMI test receiver was set with the following configurations:

| Frequency Range  | IF B/W |
|------------------|--------|
| 150 kHz – 30 MHz | 9 kHz  |

### 3.1.4 Test Procedure

The frequency and amplitude of the six highest ac power-line conducted emissions relative to the limit, measured over all the current-carrying conductors of the EUT power cords, and the operating frequency or frequency to which the EUT is tuned (if appropriate), should be reported, unless such emissions are more than 20 dB below the limit. AC power-line conducted emissions measurements are to be separately carried out only on each of the phase (“hot”) line(s) and (if used) on the neutral line(s), but not on the ground [protective earth] line(s). If less than six emission frequencies are within 20 dB of the limit, then the noise level of the measuring instrument at representative frequencies should be reported. The specific conductor of the power-line cord for each of the reported emissions should be identified. Measure the six highest emissions with respect to the limit on each current-carrying conductor of each power cord associated with the EUT (but not the power cords of associated or peripheral equipment that are part of the test configuration). Then, report the six highest emissions with respect to the limit from among all the measurements identifying the frequency and specific current-carrying conductor identified with the emission. The six highest emissions should be reported for each of the current-carrying conductors, or the six highest emissions may be reported over all the current-carrying conductors.

### 3.1.5 Corrected Amplitude & Margin Calculation

The basic equation is as follows:

Result = Reading + Factor

Factor = attenuation caused by cable loss + voltage division factor of AMN

The “**Margin**” column of the following data tables indicates the degree of compliance within the applicable limit. The equation for margin calculation is as follows:

Margin = Limit – Result

## 3.2 Radiation Spurious Emissions

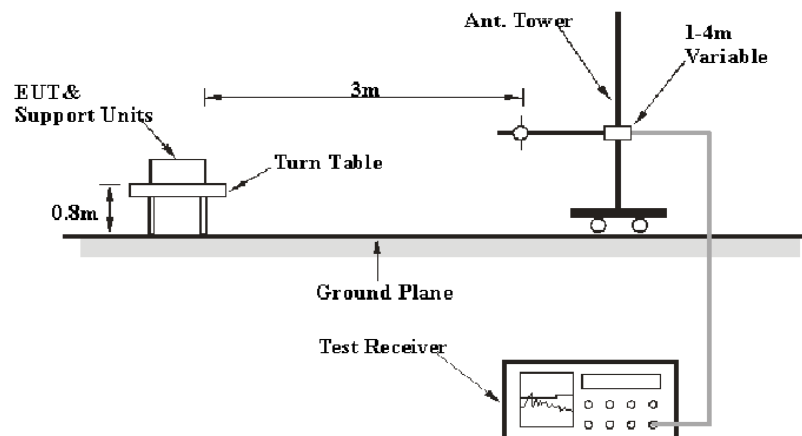
### 3.2.1 Applicable Standard

FCC §15.247 (d);

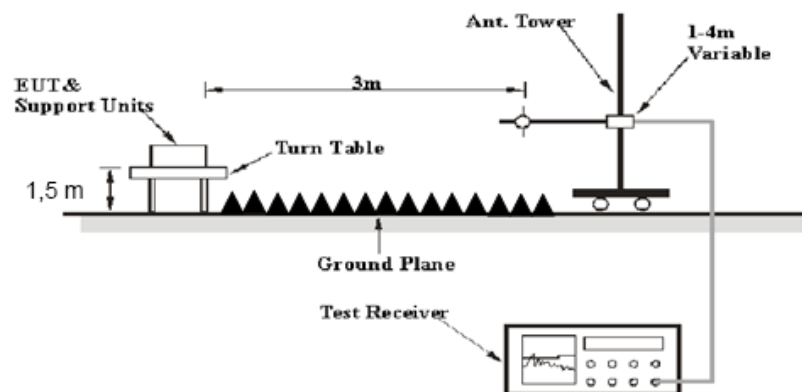
In any 100 kHz bandwidth outside the frequency band in which the spread spectrum or digitally modulated intentional radiator is operating, the radio frequency power that is produced by the intentional radiator shall be at least 20 dB below that in the 100 kHz bandwidth within the band that contains the highest level of the desired power, based on either an RF conducted or a radiated measurement, provided the transmitter demonstrates compliance with the peak conducted power limits. If the transmitter complies with the conducted power limits based on the use of RMS averaging over a time interval, as permitted under paragraph (b)(3) of this section, the attenuation required under this paragraph shall be 30 dB instead of 20 dB. Attenuation below the general limits specified in §15.209(a) is not required. In addition, radiated emissions which fall in the restricted bands, as defined in §15.205(a), must also comply with the radiated emission limits specified in §15.209(a) (see §15.205(c)).

### 3.2.2 EUT Setup

**Below 1GHz:**



**Above 1GHz:**



The radiated emissions were performed in the 3 meters distance, using the setup accordance with the [ANSI C63.10-2020](#). The specification used was the FCC 15.209, and FCC 15.247 limits.

The external I/O cables were draped along the test table and formed a bundle 30 to 40 cm long in the middle.

The spacing between the peripherals was 10 cm.

### 3.2.3 EMI Test Receiver & Spectrum Analyzer Setup

The system was investigated from 30 MHz to 10 GHz.

During the radiated emission test, the EMI test receiver & Spectrum Analyzer Setup were set with the following configurations:

30MHz-1000MHz:

| Measurement | RBW     | Video B/W | IF B/W |
|-------------|---------|-----------|--------|
| QP          | 120 kHz | 300 kHz   | 120kHz |

1GHz- 25GHz:

| Measurement | Duty cycle | RBW  | Video B/W  |
|-------------|------------|------|------------|
| PK          | Any        | 1MHz | 3 MHz      |
| AV          | >98%       | 1MHz | 10 Hz      |
|             | <98%       | 1MHz | $\geq 1/T$ |

Note: T is minimum transmission duration

If the maximized peak measured value complies with under the QP/Average limit more than 6dB, then it is unnecessary to perform an QP/Average measurement.

### 3.2.4 Test Procedure

Maximizing procedure was performed on the highest emissions to ensure that the EUT complied with all installation combinations.

Data was recorded in Quasi-peak detection mode for frequency range of 30 MHz-1 GHz, peak and Average detection modes for frequencies above 1 GHz.

### 3.2.5 Corrected Amplitude & Margin Calculation

The basic equation is as follows:

Result = Reading + Factor

Factor = Antenna Factor + Cable Loss- Amplifier Gain

The “**Margin**” column of the following data tables indicates the degree of compliance within the applicable limit. The equation for margin calculation is as follows:

Margin = Limit – Result

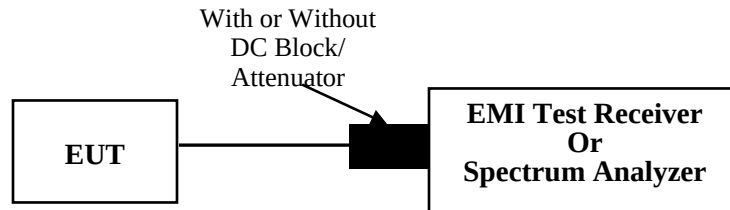
### 3.3 6 dB Emission Bandwidth:

#### 3.3.1 Applicable Standard

FCC §15.247 (a)(2)

Systems using digital modulation techniques may operate in the 902-928 MHz, 2400-2483.5 MHz, and 5725-5850 MHz bands. The minimum 6 dB bandwidth shall be at least 500 kHz.

#### 3.3.2 EUT Setup



#### 3.3.3 Test Procedure

According to **ANSI C63.10-2020** Section 11.8

- a) Set RBW = 100 kHz.
- b) Set the video bandwidth (VBW)  $\geq 3 \times \text{RBW}$ .
- c) Detector = Peak.
- d) Trace mode = max hold.
- e) Sweep = auto couple.
- f) Allow the trace to stabilize.
- g) Measure the maximum width of the emission that is constrained by the frequencies associated with the two outermost amplitude points (upper and lower frequencies) that are attenuated by 6 dB relative to the maximum level measured in the fundamental emission.

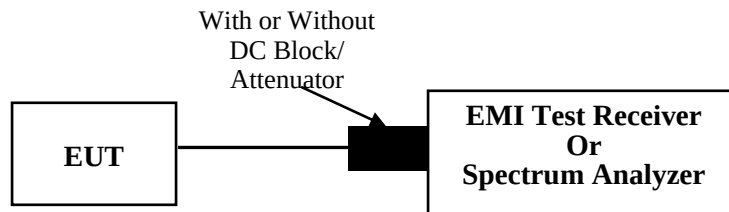
### 3.4 Maximum Conducted Output Power:

#### 3.4.1 Applicable Standard

FCC §15.247 (b)(3)

For systems using digital modulation in the 902-928 MHz, 2400-2483.5 MHz, and 5725-5850 MHz bands: 1 Watt. As an alternative to a peak power measurement, compliance with the one Watt limit can be based on a measurement of the maximum conducted output power. Maximum Conducted Output Power is defined as the total transmit power delivered to all antennas and antenna elements averaged across all symbols in the signaling alphabet when the transmitter is operating at its maximum power control level. Power must be summed across all antennas and antenna elements. The average must not include any time intervals during which the transmitter is off or is transmitting at a reduced power level. If multiple modes of operation are possible (e.g., alternative modulation methods), the maximum conducted output power is the highest total transmit power occurring in any mode.

#### 3.4.2 EUT Setup



#### 3.4.3 Test Procedure

According to **ANSI C63.10-2020** Section 11.9.1.1

The following procedure shall be used when an instrument with a resolution bandwidth that is greater than the DTS bandwidth is available to perform the measurement:

- a) Set the RBW  $\geq$  DTS bandwidth.
- b) Set VBW  $\geq$   $[3 \times \text{RBW}]$ .
- c) Set span  $\geq$   $[3 \times \text{RBW}]$ .
- d) Sweep time = auto couple.
- e) Detector = peak.
- f) Trace mode = max hold.
- g) Allow trace to fully stabilize.
- h) Use peak marker function to determine the peak amplitude level.

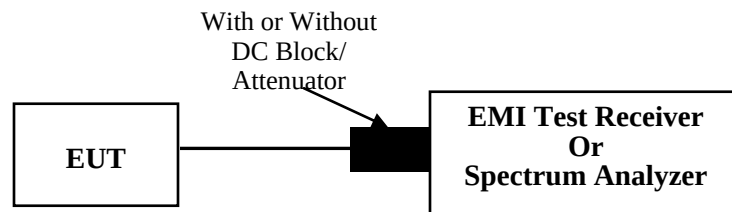
### 3.5 Maximum power spectral density:

#### 3.5.1 Applicable Standard

FCC §15.247 (e)

For digitally modulated systems, the power spectral density conducted from the intentional radiator to the antenna shall not be greater than 8 dBm in any 3 kHz band during any time interval of continuous transmission. This power spectral density shall be determined in accordance with the provisions of paragraph (b) of this section. The same method of determining the conducted output power shall be used to determine the power spectral density.

#### 3.5.2 EUT Setup



#### 3.5.3 Test Procedure

According to **ANSI C63.10-2020** Section 11.10.2

- Set analyzer center frequency to DTS channel center frequency.
- Set the span to 1.5 times the DTS bandwidth.
- Set the RBW to  $3 \text{ kHz} \leq \text{RBW} \leq 100 \text{ kHz}$ .
- Set the VBW  $\geq [3 \times \text{RBW}]$ .
- Detector = peak.
- Sweep time = auto couple.
- Trace mode = max hold.
- Allow trace to fully stabilize.
- Use the peak marker function to determine the maximum amplitude level within the RBW.
- If measured value exceeds requirement, then reduce RBW (but no less than 3 kHz) and repeat.

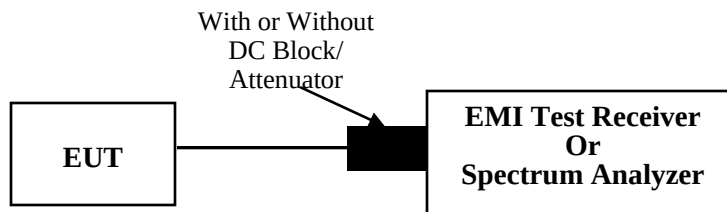
### 3.6 100 kHz Bandwidth of Frequency Band Edge:

#### 3.6.1 Applicable Standard

FCC §15.247 (d);

In any 100 kHz bandwidth outside the frequency band in which the spread spectrum or digitally modulated intentional radiator is operating, the radio frequency power that is produced by the intentional radiator shall be at least 20 dB below that in the 100 kHz bandwidth within the band that contains the highest level of the desired power, based on either an RF conducted or a radiated measurement, provided the transmitter demonstrates compliance with the peak conducted power limits. If the transmitter complies with the conducted power limits based on the use of RMS averaging over a time interval, as permitted under paragraph (b)(3) of this section, the attenuation required under this paragraph shall be 30 dB instead of 20 dB. Attenuation below the general limits specified in §15.209(a) is not required. In addition, radiated emissions which fall in the restricted bands, as defined in §15.205(a), must also comply with the radiated emission limits specified in §15.209(a) (see §15.205(c)).

#### 3.6.2 EUT Setup



#### 3.6.3 Test Procedure

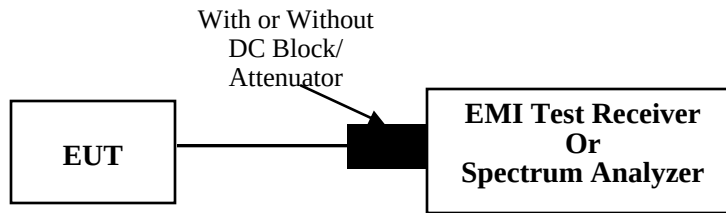
According to **ANSI C63.10-2020** Section 11.11

- a) Set the center frequency and span to encompass frequency range to be measured.
- b) Set the RBW = 100 kHz.
- c) Set the VBW  $\geq [3 \times \text{RBW}]$ .
- d) Detector = peak.
- e) Sweep time = auto couple.
- f) Trace mode = max hold.
- g) Allow trace to fully stabilize.
- h) Use the peak marker function to determine the maximum amplitude level.

Ensure that the amplitude of all unwanted emissions outside of the authorized frequency band (excluding restricted frequency bands) is attenuated by at least the minimum requirements specified in 11.11. Report the three highest emissions relative to the limit.

### 3.7 Duty Cycle:

#### 3.7.1 EUT Setup



#### 3.7.2 Test Procedure

According to **ANSI C63.10-2020** Section 11.6

The zero-span mode on a spectrum analyzer or EMI receiver if the response time and spacing between bins on the sweep are sufficient to permit accurate measurements of the ON and OFF times of the transmitted signal:

- 1) Set the center frequency of the instrument to the center frequency of the transmission.
- 2) Set  $RBW \geq OBW$  if possible; otherwise, set RBW to the largest available value.
- 3) Set  $VBW \geq RBW$ . Set detector = peak or average.
- 4) The zero-span measurement method shall not be used unless both RBW and VBW are  $> 50/T$  and the number of sweep points across duration T exceeds 100. (For example, if VBW and/or RBW are limited to 3 MHz, then the zero-span method of measuring the duty cycle shall not be used if  $T \leq 16.7 \mu s$ .)

### 3.8 Antenna Requirement

#### 3.8.1 Applicable Standard

FCC §15.203

An intentional radiator shall be designed to ensure that no antenna other than that furnished by the responsible party shall be used with the device. The use of a permanently attached antenna or of an antenna that uses a unique coupling to the intentional radiator shall be considered sufficient to comply with the provisions of this section. The manufacturer may design the unit so that a broken antenna can be replaced by the user, but the use of a standard antenna jack or electrical connector is prohibited. This requirement does not apply to carrier current devices or to devices operated under the provisions of §§15.211, 15.213, 15.217, 15.219, 15.221, or §15.236. Further, this requirement does not apply to intentional radiators that must be professionally installed, such as perimeter protection systems and some field disturbance sensors, or to other intentional radiators which, in accordance with §15.31(d), must be measured at the installation site. However, the installer shall be responsible for ensuring that the proper antenna is employed so that the limits in this part are not exceeded.

#### 3.8.2 Judgment

**Compliant.** Please refer to the Antenna Information detail in Section 1.

## 4. Test DATA AND RESULTS

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### 4.1 AC Line Conducted Emissions

**Not Applicable**, the device was powered by battery.

## 4.2 Radiation Spurious Emissions

|                |                      |              |                   |
|----------------|----------------------|--------------|-------------------|
| Serial Number: | 299I-5               | Test Date:   | 2023/9/4~2023/9/9 |
| Test Site:     | 966-1, 966-2         | Test Mode:   | Transmitting      |
| Tester:        | Hugo Huo, Mack Huang | Test Result: | Pass              |

| Environmental Conditions: |           |                           |       |                           |            |
|---------------------------|-----------|---------------------------|-------|---------------------------|------------|
| Temperature:<br>(°C)      | 24.1~26.3 | Relative Humidity:<br>(%) | 58~59 | ATM<br>Pressure:<br>(kPa) | 99.6~100.1 |

### Test Equipment List and Details:

| Manufacturer                  | Description       | Model                 | Serial Number | Calibration Date | Calibration Due Date |
|-------------------------------|-------------------|-----------------------|---------------|------------------|----------------------|
| Radiated emissions below 1GHz |                   |                       |               |                  |                      |
| Sunol Sciences                | Antenna           | JB6                   | A082520-5     | 2020/10/19       | 2023/10/18           |
| R&S                           | EMI Test Receiver | ESR3                  | 102724        | 2023/3/31        | 2024/3/30            |
| TIMES MICROWAVE               | Coaxial Cable     | LMR-600-UltraFlex     | C-0470-02     | 2023/7/16        | 2024/7/15            |
| TIMES MICROWAVE               | Coaxial Cable     | LMR-600-UltraFlex     | C-0780-01     | 2023/7/16        | 2024/7/15            |
| Sonoma                        | Amplifier         | 310N                  | 186165        | 2023/7/16        | 2024/7/15            |
| Audix                         | Test Software     | E3                    | 201021 (V9)   | N/A              | N/A                  |
| Radiated emissions above 1GHz |                   |                       |               |                  |                      |
| ETS-Lindgren                  | Horn Antenna      | 3115                  | 9912-5985     | 2020/10/13       | 2023/10/12           |
| R&S                           | Spectrum Analyzer | FSV40                 | 101591        | 2023/3/31        | 2024/3/30            |
| MICRO-COAX                    | Coaxial Cable     | UFA210A-1-1200-70U300 | 217423-008    | 2023/8/6         | 2024/8/5             |
| MICRO-COAX                    | Coaxial Cable     | UFA210A-1-2362-300300 | 235780-001    | 2023/8/6         | 2024/8/5             |
| Mini                          | Pre-amplifier     | ZVA-183-S+            | 5969001149    | 2022/11/9        | 2023/11/8            |
| Audix                         | Test Software     | E3                    | 201021 (V9)   | N/A              | N/A                  |

**\* Statement of Traceability:** China Certification ICT Co., Ltd (Dongguan) attests that all calibrations have been performed, traceable to National Primary Standards and International System of Units (SI).

### Test Data:

Please refer to the below table and plots.

After pre-scan in the X, Y and Z axes of orientation, the worst case is below:

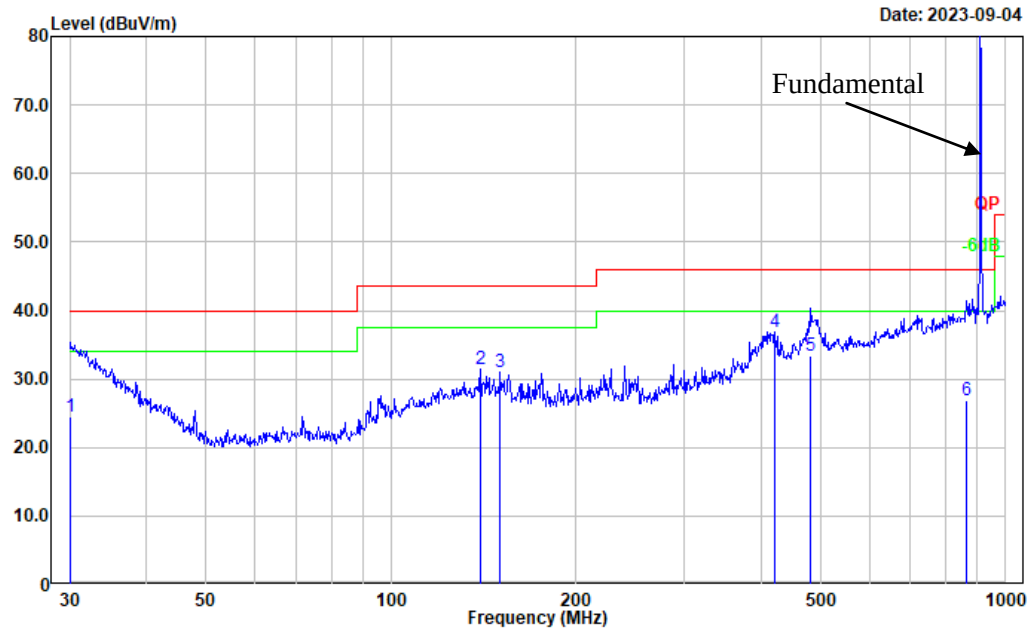
## 1) 30MHz-1GHz (Low channel was the worst)

Project No.: CR230844753-RF

Tester: Hugo Huo

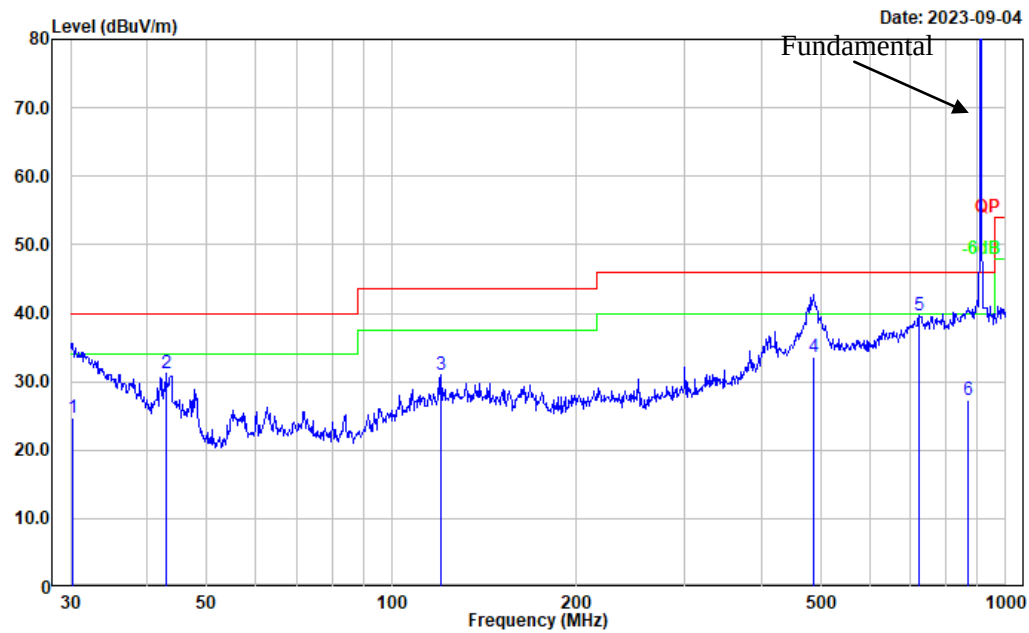
Polarization: horizontal

Note:



| No. | Frequency (MHz) | Reading (dBμV) | Factor (dB/m) | Result (dBμV/m) | Limit (dBμV/m) | Margin (dB) | Detector |
|-----|-----------------|----------------|---------------|-----------------|----------------|-------------|----------|
| 1   | 30.000          | -3.37          | 27.88         | 24.51           | 40.00          | 15.49       | QP       |
| 2   | 139.851         | 11.86          | 19.66         | 31.52           | 43.50          | 11.98       | Peak     |
| 3   | 150.538         | 11.51          | 19.40         | 30.91           | 43.50          | 12.59       | Peak     |
| 4   | 422.058         | 13.54          | 23.37         | 36.91           | 46.00          | 9.09        | Peak     |
| 5   | 482.216         | 8.39           | 25.00         | 33.39           | 46.00          | 12.61       | QP       |
| 6   | 863.056         | -2.81          | 29.60         | 26.79           | 46.00          | 19.21       | QP       |

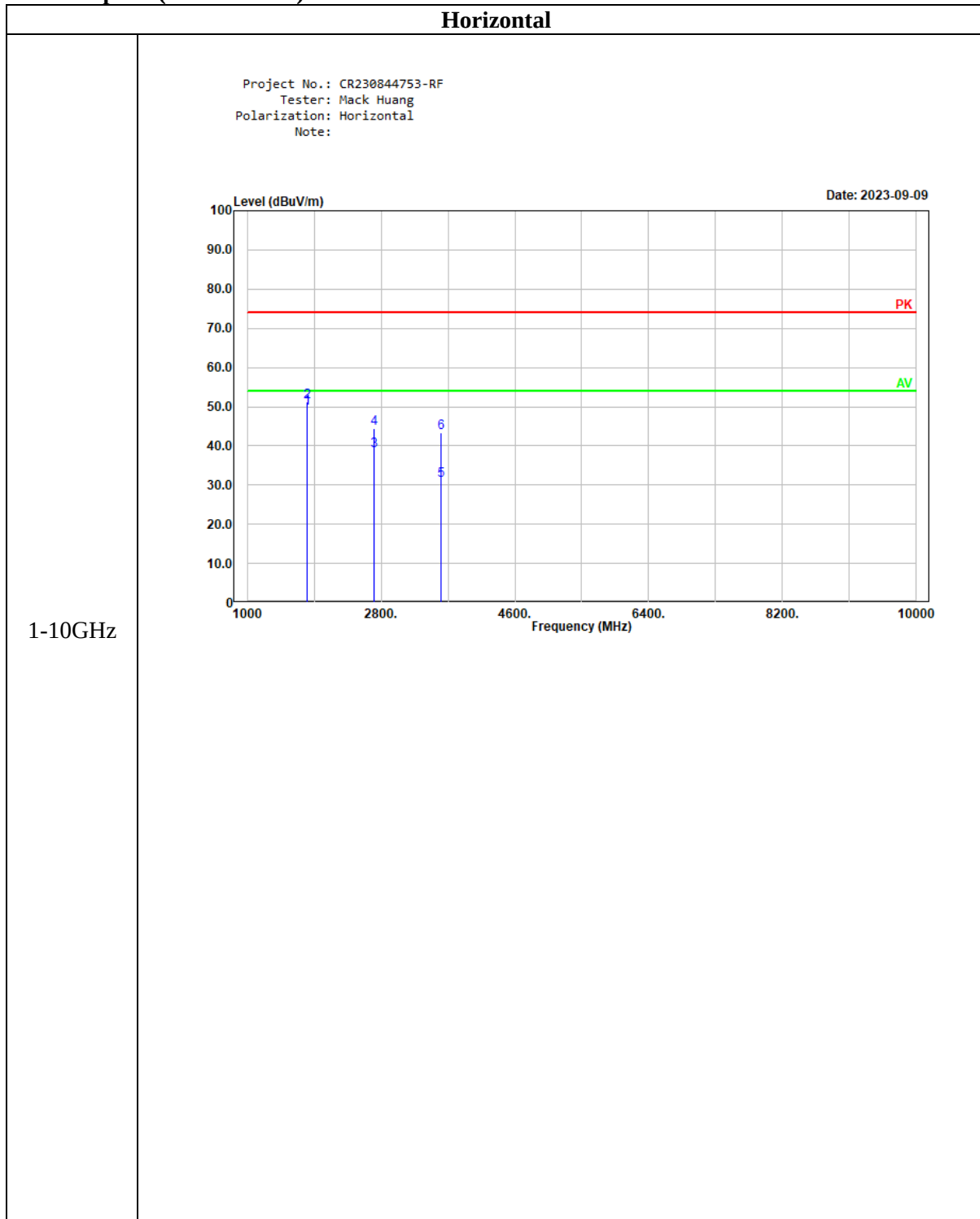
Project No.: CR230844753-RF  
Tester: Hugo Huo  
Polarization: vertical  
Note:



| No. | Frequency<br>(MHz) | Reading<br>(dBμV) | Factor<br>(dB/m) | Result<br>(dBμV/m) | Limit<br>(dBμV/m) | Margin<br>(dB) | Detector |
|-----|--------------------|-------------------|------------------|--------------------|-------------------|----------------|----------|
| 1   | 30.317             | -2.92             | 27.63            | 24.71              | 40.00             | 15.29          | QP       |
| 2   | 42.900             | 12.64             | 18.49            | 31.13              | 40.00             | 8.87           | Peak     |
| 3   | 120.277            | 11.02             | 19.99            | 31.01              | 43.50             | 12.49          | Peak     |
| 4   | 487.315            | 8.60              | 25.05            | 33.65              | 46.00             | 12.35          | QP       |
| 5   | 721.726            | 11.59             | 28.04            | 39.63              | 46.00             | 6.37           | Peak     |
| 6   | 866.088            | -2.26             | 29.59            | 27.33              | 46.00             | 18.67          | QP       |

## 2) 1-10GHz:

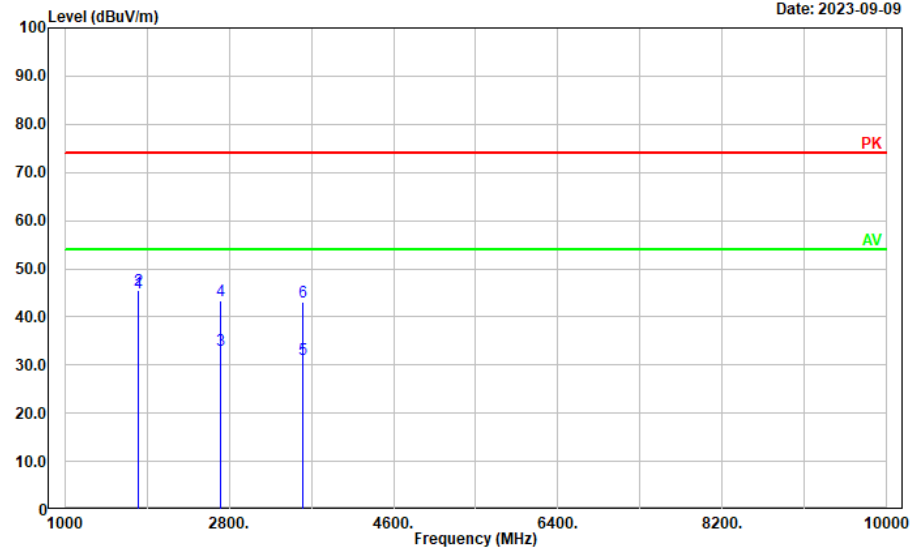
| 1-10GHz.                  |                   |          |                |                  |                    |                   |                |
|---------------------------|-------------------|----------|----------------|------------------|--------------------|-------------------|----------------|
| Frequency<br>(MHz)        | Receiver          |          | Polar<br>(H/V) | Factor<br>(dB/m) | Result<br>(dBµV/m) | Limit<br>(dBµV/m) | Margin<br>(dB) |
|                           | Reading<br>(dBµV) | Detector |                |                  |                    |                   |                |
| Low Channel:903MHz        |                   |          |                |                  |                    |                   |                |
| 1806.000                  | 50.70             | PK       | H              | 1.32             | 52.02              | 74.00             | 21.98          |
| 1806.000                  | 49.44             | AV       | H              | 1.32             | 50.76              | 54.00             | 3.24           |
| 1806.000                  | 45.94             | PK       | V              | 1.32             | 47.26              | 74.00             | 26.74          |
| 1806.000                  | 44.00             | AV       | V              | 1.32             | 45.32              | 54.00             | 8.68           |
| 2709.000                  | 40.78             | PK       | H              | 4.76             | 45.54              | 74.00             | 28.46          |
| 2709.000                  | 35.70             | AV       | H              | 4.76             | 40.46              | 54.00             | 13.54          |
| 2709.000                  | 39.63             | PK       | V              | 4.76             | 44.39              | 74.00             | 29.61          |
| 2709.000                  | 29.88             | AV       | V              | 4.76             | 34.64              | 54.00             | 19.36          |
| 3612.000                  | 36.28             | PK       | H              | 8.00             | 44.28              | 74.00             | 29.72          |
| 3612.000                  | 24.33             | AV       | H              | 8.00             | 32.33              | 54.00             | 21.67          |
| 3612.000                  | 36.15             | PK       | V              | 8.00             | 44.15              | 74.00             | 29.85          |
| 3612.000                  | 24.18             | AV       | V              | 8.00             | 32.18              | 54.00             | 21.82          |
| Middle Channel: 907.8 MHz |                   |          |                |                  |                    |                   |                |
| 1815.600                  | 49.03             | PK       | H              | 1.36             | 50.39              | 74.00             | 23.61          |
| 1815.600                  | 47.47             | AV       | H              | 1.36             | 48.83              | 54.00             | 5.17           |
| 1815.600                  | 45.31             | PK       | V              | 1.36             | 46.67              | 74.00             | 27.33          |
| 1815.600                  | 40.73             | AV       | V              | 1.36             | 42.09              | 54.00             | 11.91          |
| 2723.400                  | 41.32             | PK       | H              | 4.82             | 46.14              | 74.00             | 27.86          |
| 2723.400                  | 36.61             | AV       | H              | 4.82             | 41.43              | 54.00             | 12.57          |
| 2723.400                  | 40.54             | PK       | V              | 4.82             | 45.36              | 74.00             | 28.64          |
| 2723.400                  | 32.48             | AV       | V              | 4.82             | 37.3               | 54.00             | 16.70          |
| 3631.200                  | 35.91             | PK       | H              | 8.03             | 43.94              | 74.00             | 30.06          |
| 3631.200                  | 23.47             | AV       | H              | 8.03             | 31.5               | 54.00             | 22.50          |
| 3631.200                  | 36.50             | PK       | V              | 8.03             | 44.53              | 74.00             | 29.47          |
| 3631.200                  | 23.20             | AV       | V              | 8.03             | 31.23              | 54.00             | 22.77          |
| High Channel: 914.2 MHz   |                   |          |                |                  |                    |                   |                |
| 1828.400                  | 49.51             | PK       | H              | 1.43             | 50.94              | 74.00             | 23.06          |
| 1828.400                  | 49.28             | AV       | H              | 1.43             | 50.71              | 54.00             | 3.29           |
| 1828.400                  | 47.04             | PK       | V              | 1.43             | 48.47              | 74.00             | 25.53          |
| 1828.400                  | 44.90             | AV       | V              | 1.43             | 46.33              | 54.00             | 7.67           |
| 2742.600                  | 43.23             | PK       | H              | 4.90             | 48.13              | 74.00             | 25.87          |
| 2742.600                  | 38.87             | AV       | H              | 4.90             | 43.77              | 54.00             | 10.23          |
| 2742.600                  | 41.59             | PK       | V              | 4.90             | 46.49              | 74.00             | 27.51          |
| 2742.600                  | 33.97             | AV       | V              | 4.90             | 38.87              | 54.00             | 15.13          |
| 3656.800                  | 35.58             | PK       | H              | 8.10             | 43.68              | 74.00             | 30.32          |
| 3656.800                  | 22.14             | AV       | H              | 8.10             | 30.24              | 54.00             | 23.76          |
| 3656.800                  | 34.34             | PK       | V              | 8.10             | 42.44              | 74.00             | 31.56          |
| 3656.800                  | 21.28             | AV       | V              | 8.10             | 29.38              | 54.00             | 24.62          |

**Worst Test plots (Low channel)**

**Vertical**

Project No.: CR230844753-RF  
Tester: Mack Huang  
Polarization: Vertical  
Note:

Date: 2023-09-09



1-10GHz

**4.3 6 dB Emission Bandwidth:**

|                |           |              |              |
|----------------|-----------|--------------|--------------|
| Serial Number: | 299I-5    | Test Date:   | 2023/8/4     |
| Test Site:     | RF        | Test Mode:   | Transmitting |
| Tester:        | Panda Sun | Test Result: | Pass         |

**Environmental Conditions:**

|                      |      |                           |    |                        |     |
|----------------------|------|---------------------------|----|------------------------|-----|
| Temperature:<br>(°C) | 25.3 | Relative Humidity:<br>(%) | 56 | ATM Pressure:<br>(kPa) | 101 |
|----------------------|------|---------------------------|----|------------------------|-----|

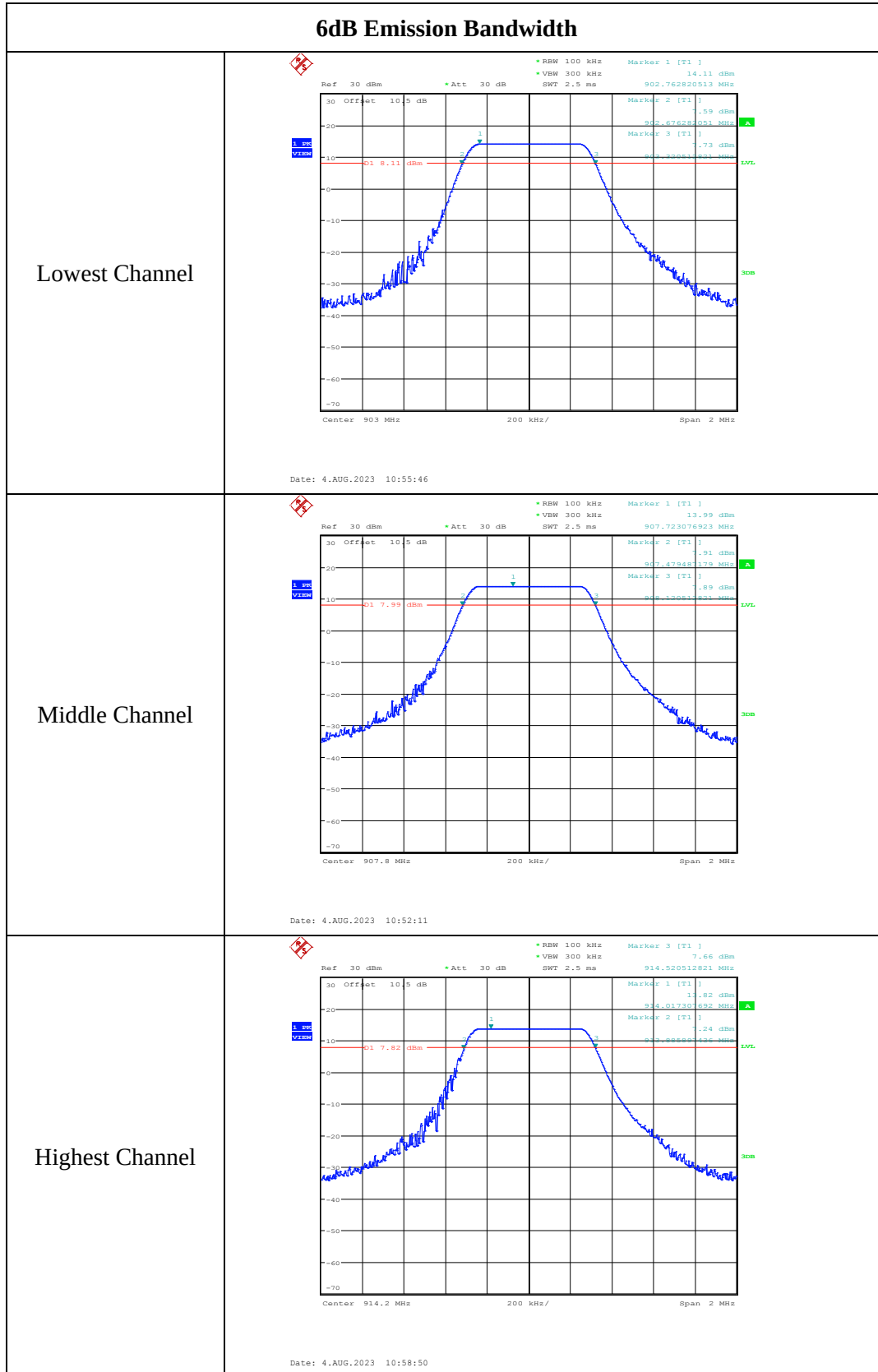
**Test Equipment List and Details:**

| Manufacturer | Description        | Model         | Serial Number | Calibration Date | Calibration Due Date |
|--------------|--------------------|---------------|---------------|------------------|----------------------|
| R&S          | Spectrum Analyzer  | FSU26         | 200120        | 2023/4/18        | 2024/4/17            |
| zhuoxiang    | Coaxial Cable      | SMA-178       | 211001        | Each time        | N/A                  |
| eastsheep    | Coaxial Attenuator | 2W-SMA-JK-18G | 21060301      | Each time        | N/A                  |

*\* Statement of Traceability: China Certification ICT Co., Ltd (Dongguan) attests that all calibrations have been performed, traceable to National Primary Standards and International System of Units (SI).*

**Test Data:**

| Test Channel | Test Frequency (MHz) | 6 dB Bandwidth (MHz) | Limit (MHz) |
|--------------|----------------------|----------------------|-------------|
| Lowest       | 903                  | 0.644                | $\geq 0.5$  |
| Middle       | 907.8                | 0.641                | $\geq 0.5$  |
| Highest      | 914.2                | 0.635                | $\geq 0.5$  |



**4.4 Maximum Conducted Output Power:**

|                |           |              |              |
|----------------|-----------|--------------|--------------|
| Serial Number: | 299I-5    | Test Date:   | 2023/8/4     |
| Test Site:     | RF        | Test Mode:   | Transmitting |
| Tester:        | Panda Sun | Test Result: | Pass         |

**Environmental Conditions:**

|                      |      |                              |    |                        |     |
|----------------------|------|------------------------------|----|------------------------|-----|
| Temperature:<br>(°C) | 25.3 | Relative<br>Humidity:<br>(%) | 56 | ATM Pressure:<br>(kPa) | 101 |
|----------------------|------|------------------------------|----|------------------------|-----|

**Test Equipment List and Details:**

| Manufacturer | Description        | Model         | Serial Number | Calibration Date | Calibration Due Date |
|--------------|--------------------|---------------|---------------|------------------|----------------------|
| R&S          | Spectrum Analyzer  | FSU26         | 200120        | 2023/4/18        | 2024/4/17            |
| zhuoxiang    | Coaxial Cable      | SMA-178       | 211001        | Each time        | N/A                  |
| eastsheep    | Coaxial Attenuator | 2W-SMA-JK-18G | 21060301      | Each time        | N/A                  |

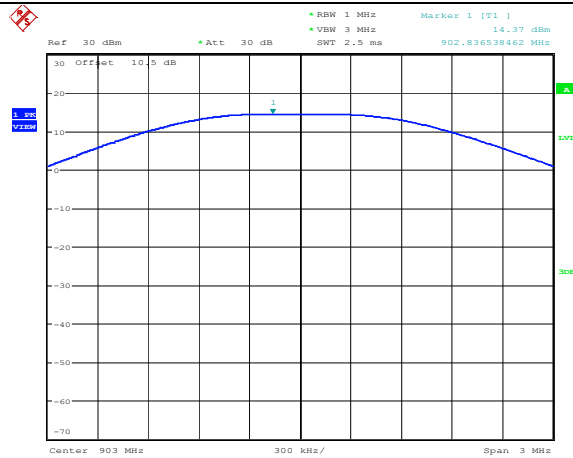
*\* Statement of Traceability: China Certification ICT Co., Ltd (Dongguan) attests that all calibrations have been performed, traceable to National Primary Standards and International System of Units (SI).*

**Test Data:**

| Test Channel | Test Frequency (MHz) | Maximum Conducted Peak Output Power (dBm) | Limit (dBm) |
|--------------|----------------------|-------------------------------------------|-------------|
| Lowest       | 903                  | 14.37                                     | ≤30         |
| Middle       | 907.8                | 14.19                                     | ≤30         |
| Highest      | 914.2                | 13.97                                     | ≤30         |

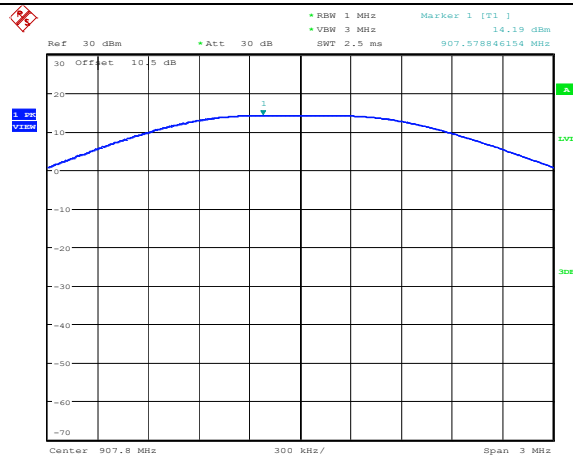
## Maximum Conducted Output Power

Lowest



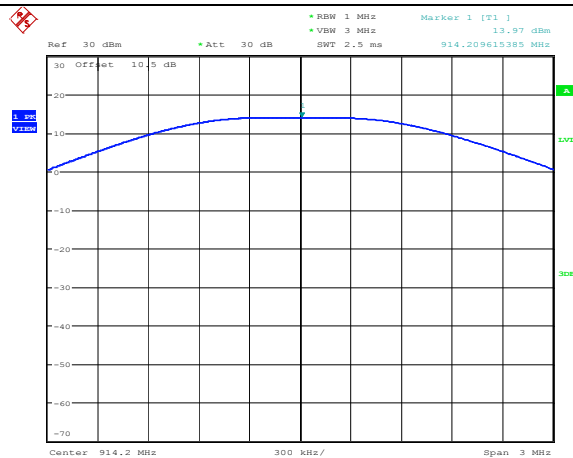
Date: 4.AUG.2023 10:30:15

Middle



Date: 4.AUG.2023 10:26:55

Highest



Date: 4.AUG.2023 10:24:39

**4.5 Maximum power spectral density:**

|                |           |              |              |
|----------------|-----------|--------------|--------------|
| Serial Number: | 299I-5    | Test Date:   | 2023/8/4     |
| Test Site:     | RF        | Test Mode:   | Transmitting |
| Tester:        | Panda Sun | Test Result: | Pass         |

**Environmental Conditions:**

|                      |      |                           |    |                        |     |
|----------------------|------|---------------------------|----|------------------------|-----|
| Temperature:<br>(°C) | 25.3 | Relative Humidity:<br>(%) | 56 | ATM Pressure:<br>(kPa) | 101 |
|----------------------|------|---------------------------|----|------------------------|-----|

**Test Equipment List and Details:**

| Manufacturer | Description        | Model         | Serial Number | Calibration Date | Calibration Due Date |
|--------------|--------------------|---------------|---------------|------------------|----------------------|
| R&S          | Spectrum Analyzer  | FSU26         | 200120        | 2023/4/18        | 2024/4/17            |
| zhuoxiang    | Coaxial Cable      | SMA-178       | 211001        | Each time        | N/A                  |
| eastsheep    | Coaxial Attenuator | 2W-SMA-JK-18G | 21060301      | Each time        | N/A                  |

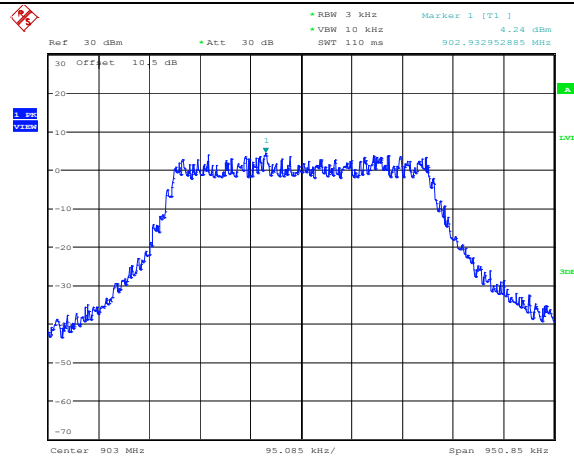
*\* Statement of Traceability: China Certification ICT Co., Ltd (Dongguan) attests that all calibrations have been performed, traceable to National Primary Standards and International System of Units (SI).*

**Test Data:**

| Test Channel | Test Frequency<br>(MHz) | Power Spectral<br>Density<br>(dBm/3kHz) | Limit<br>(dBm/3kHz) |
|--------------|-------------------------|-----------------------------------------|---------------------|
| Lowest       | 903                     | 4.24                                    | ≤8.00               |
| Middle       | 907.8                   | 4.07                                    | ≤8.00               |
| Highest      | 914.2                   | 3.94                                    | ≤8.00               |

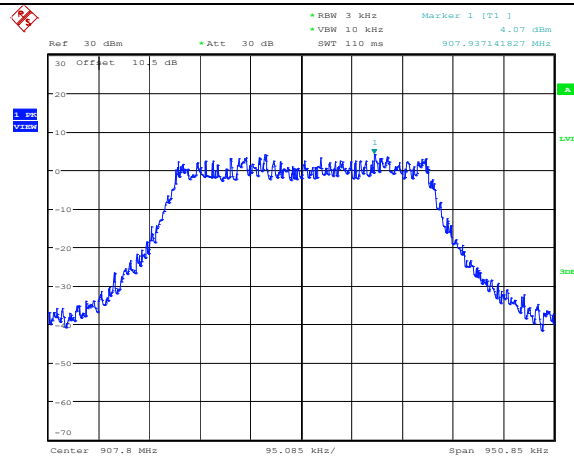
## Maximum power spectral density

Lowest Channel



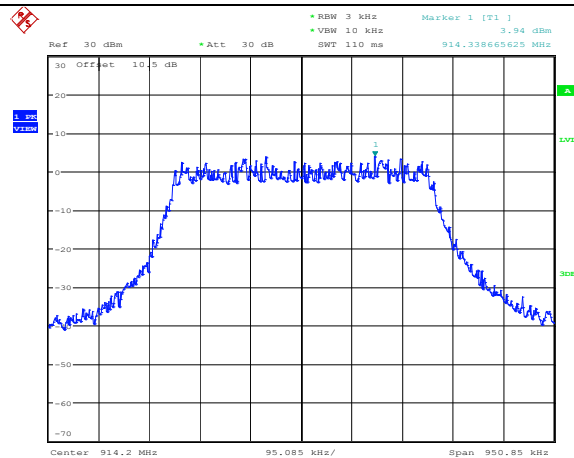
Date: 4.AUG.2023 10:17:27

Middle Channel



Date: 4.AUG.2023 10:20:51

Highest Channel



Date: 4.AUG.2023 10:23:27

**4.6 100 kHz Bandwidth of Frequency Band Edge:**

|                |           |              |              |
|----------------|-----------|--------------|--------------|
| Serial Number: | 299I-5    | Test Date:   | 2023/8/4     |
| Test Site:     | RF        | Test Mode:   | Transmitting |
| Tester:        | Panda Sun | Test Result: | Pass         |

**Environmental Conditions:**

|                      |      |                           |    |                        |     |
|----------------------|------|---------------------------|----|------------------------|-----|
| Temperature:<br>(°C) | 25.3 | Relative Humidity:<br>(%) | 56 | ATM Pressure:<br>(kPa) | 101 |
|----------------------|------|---------------------------|----|------------------------|-----|

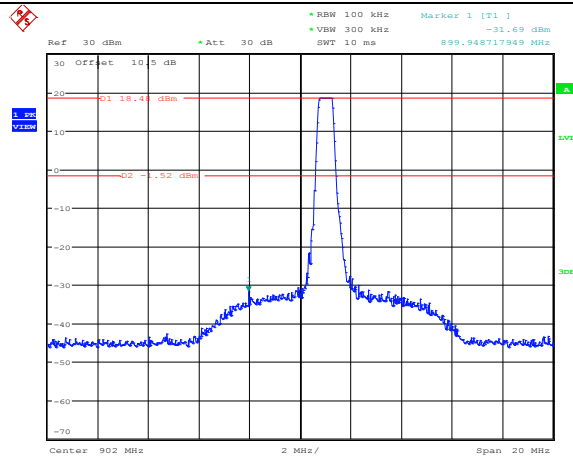
**Test Equipment List and Details:**

| Manufacturer | Description        | Model         | Serial Number | Calibration Date | Calibration Due Date |
|--------------|--------------------|---------------|---------------|------------------|----------------------|
| R&S          | Spectrum Analyzer  | FSU26         | 200120        | 2023/4/18        | 2024/4/17            |
| zhuoxiang    | Coaxial Cable      | SMA-178       | 211001        | Each time        | N/A                  |
| eastsheep    | Coaxial Attenuator | 2W-SMA-JK-18G | 21060301      | Each time        | N/A                  |

*\* Statement of Traceability: China Certification ICT Co., Ltd (Dongguan) attests that all calibrations have been performed, traceable to National Primary Standards and International System of Units (SI).*

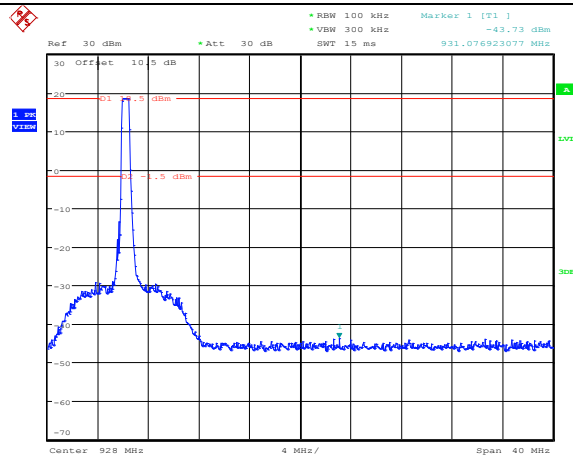
**100 kHz Bandwidth of Frequency Band Edge**

Lowest Band edge



Date: 4.AUG.2023 10:36:12

Highest Band edge



Date: 4.AUG.2023 10:41:48

**4.7 Duty Cycle:**

|                |           |              |              |
|----------------|-----------|--------------|--------------|
| Serial Number: | 299I-5    | Test Date:   | 2023/8/4     |
| Test Site:     | RF        | Test Mode:   | Transmitting |
| Tester:        | Panda Sun | Test Result: | N/A          |

**Environmental Conditions:**

|                      |      |                           |    |                        |     |
|----------------------|------|---------------------------|----|------------------------|-----|
| Temperature:<br>(°C) | 25.3 | Relative Humidity:<br>(%) | 56 | ATM Pressure:<br>(kPa) | 101 |
|----------------------|------|---------------------------|----|------------------------|-----|

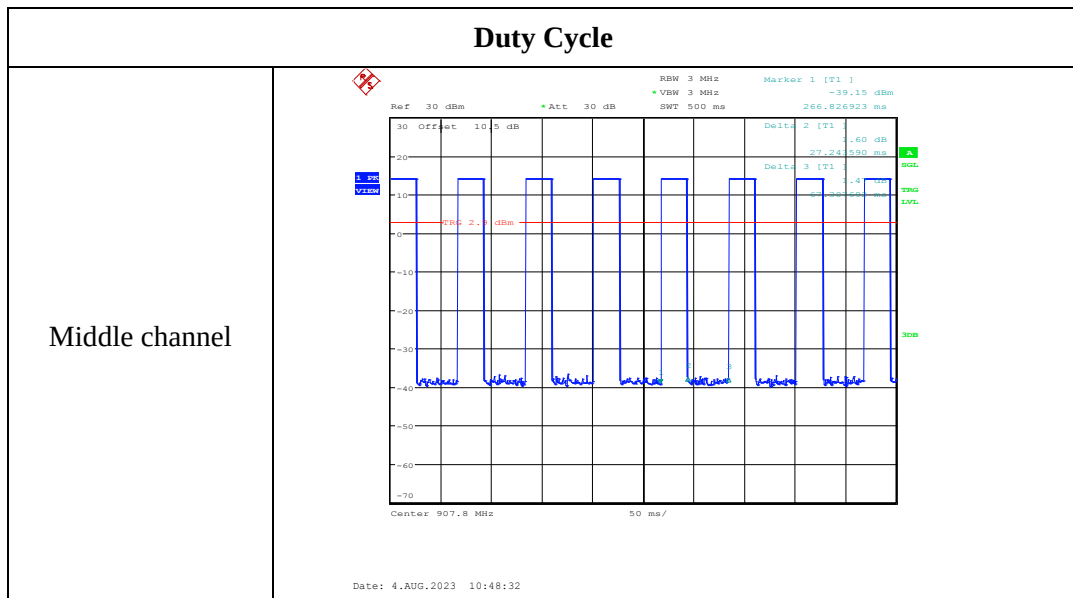
**Test Equipment List and Details:**

| Manufacturer | Description        | Model         | Serial Number | Calibration Date | Calibration Due Date |
|--------------|--------------------|---------------|---------------|------------------|----------------------|
| R&S          | Spectrum Analyzer  | FSU26         | 200120        | 2023/4/18        | 2024/4/17            |
| zhuoxiang    | Coaxial Cable      | SMA-178       | 211001        | Each time        | N/A                  |
| eastsheep    | Coaxial Attenuator | 2W-SMA-JK-18G | 21060301      | Each time        | N/A                  |

\* Statement of Traceability: China Certification ICT Co., Ltd (Dongguan) attests that all calibrations have been performed, traceable to National Primary Standards and International System of Units (SI).

**Test Data:**

| Test Channel | Test Frequency (MHz) | Ton (ms) | Ton+off (ms) | Duty Cycle (%) | 1/T (Hz) | VBW Setting (Hz) |
|--------------|----------------------|----------|--------------|----------------|----------|------------------|
| Middle       | 907.8                | 27.24    | 67.31        | 40.47          | 36.71    | 100              |



## 5. RF EXPOSURE EVALUATION

### 5.1 Applicable Standard

According to subpart §1.1310, systems operating under the provisions of this section shall be operated in a manner that ensures that the public is not exposed to radio frequency energy level in excess of the Commission's guidelines.

Limits for Maximum Permissible Exposure (MPE) (§1.1310, §2.1091)

| (B) Limits for General Population/Uncontrolled Exposure |                               |                               |                                     |                          |
|---------------------------------------------------------|-------------------------------|-------------------------------|-------------------------------------|--------------------------|
| Frequency Range (MHz)                                   | Electric Field Strength (V/m) | Magnetic Field Strength (A/m) | Power Density (mW/cm <sup>2</sup> ) | Averaging Time (minutes) |
| 0.3–1.34                                                | 614                           | 1.63                          | *(100)                              | 30                       |
| 1.34–30                                                 | 824/f                         | 2.19/f                        | *(180/f)                            | 30                       |
| 30–300                                                  | 27.5                          | 0.073                         | 0.2                                 | 30                       |
| 300–1500                                                | /                             | /                             | f/1500                              | 30                       |
| 1500–100,000                                            | /                             | /                             | 1.0                                 | 30                       |

f = frequency in MHz; \* = Plane-wave equivalent power density;

According to §1.1310 and §2.1091 RF exposure is calculated.

#### Calculation formula:

Prediction of power density at the distance of the applicable MPE limit

$S = PG/4\pi R^2$  = power density (in appropriate units, e.g. mW/cm<sup>2</sup>);

P = power input to the antenna (in appropriate units, e.g., mW);

G = power gain of the antenna in the direction of interest relative to an isotropic radiator, the power gain factor, is normally numeric gain;

R = distance to the center of radiation of the antenna (appropriate units, e.g., cm);

### 5.2 Measurement Result

| Operation Modes | Frequency (MHz) | Antenna Gain |           | Conducted output power including Tune-up Tolerance |       | Evaluation Distance (cm) | Power Density (mW/cm <sup>2</sup> ) | MPE Limit (mW/cm <sup>2</sup> ) |
|-----------------|-----------------|--------------|-----------|----------------------------------------------------|-------|--------------------------|-------------------------------------|---------------------------------|
|                 |                 | (dBi)        | (numeric) | (dBm)                                              | (mW)  |                          |                                     |                                 |
| BLE             | 2402-2480       | 2            | 1.58      | 0.5                                                | 1.12  | 20                       | 0.0004                              | 1.0                             |
| Lora- Hybrid    | 902.3-914.9     | 2            | 1.58      | 5.0                                                | 3.16  | 20                       | 0.001                               | 0.6                             |
| Lora-DTS        | 903-914.2       | 2            | 1.58      | 15.0                                               | 31.62 | 20                       | 0.010                               | 0.6                             |

Note: The Maximum Conducted Power including Tune-up Tolerance was declared by manufacturer.

The BLE, Lora- Hybrid and Lora-DTS can't transmit simultaneously.

**Result: The device compliant the MPE-Based Exemption at 20cm distances.**

## **6. EUT PHOTOGRAPHS**

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Please refer to the attachment CR230844753-EXP EUT EXTERNAL PHOTOGRAPHS and  
CR230844753-INP EUT INTERNAL PHOTOGRAPHS

## 7. TEST SETUP PHOTOGRAPHS

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Please refer to the attachment CR230844753-00C-TSP TEST SETUP PHOTOGRAPHS.

**===== END OF REPORT =====**